Class XII Session 2025-26
Subject - Mathematics
Sample Question Paper - 2

Time Allowed: 30minutes

General Instructions:

1. This Question paper contains 13 questions. All questions are compulsory.
2. This Question paper is divided into five Sections -
3. Use of calculators is not allowed.

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Section A

1. If $|A| = 2$, where A is a 2×2 matrix, then $|4A^{-1}|$ equals: **[1]**

   a) 2 b) 4

   c) 8 d) $\frac{1}{32}$

2. Let A be a 3×3 matrix such that $|\text{adj } A| = 64$. Then $|A|$ is equal to: **[1]**

   a) 8 only b) 64

   c) 8 or -8 d) -8 only

3. Let A be the area of a triangle having vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3). Which of the following is correct? **[1]**

   a) $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = A^2$

   b) $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \pm 2A$

   c) $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \pm \frac{A}{2}$

   d) $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \pm A$

4. If $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$, then the value of $(2x + y - z)$ is: **[1]**

   a) 2 b) 5

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c) 1 d) 3

5. The equation of a line passing through point (2, -1, 0) and parallel to the line $\frac{x}{1} = \frac{y-1}{2} = \frac{2-z}{2}$ is: **[1]**

a) $\frac{x-2}{1} = \frac{y-1}{2} = \frac{z}{2}$ b) $\frac{x-2}{1} = \frac{y+1}{2} = \frac{z}{-2}$

c) $\frac{x+2}{1} = \frac{y-1}{2} = \frac{z}{-2}$ d) $\frac{x+2}{1} = \frac{y-1}{2} = \frac{z}{2}$

6. The solution of the DE cos x(1 + cos y)dx - sin y(1 + sin x)dy = 0 is **[1]**

a) 1 + sin x cos y = C b) 2sin x cos y - cos x = C

c) sin x cos y + cos x = C d) (1 + sin x)(1 + cos y) = C

7. A linear programming problem deals with the optimization of a/an: **[1]**

a) linear function b) logarithmic function

c) quadratic function d) exponential function

8. The value of p for which the vectors $2i + pj + k$ and $-4i - 6j + 26k$ are perpendicular to each other, is: **[1]**

a) 3 b) -3

c) $-\frac{17}{3}$ d) $\frac{17}{3}$

9. $\int x^2 e^{x^3} dx$ equals **[1]**

a) $\frac{1}{2}e^{x^2} + C$ b) $\frac{1}{2}e^{x^3} + C$

c) $\frac{1}{3}e^{x^3} + C$ d) $\frac{1}{3}e^{x^2} + C$

10. Find the value of x and y are respectively, if $\begin{bmatrix} x & y \\ 3y & x \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$. **[1]**

a) 3, 2 b) 4, 3

c) 2, 1 d) 1, 1

11. Corner points of the feasible region for an LPP are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5). Let F = 4x + 6y be the objective function. Maximum of F – Minimum of F = **[1]**

a) 18 b) 60

c) 42 d) 48

12. For any two vectors $\vec{a}$ and $\vec{b}$ which of the following statements is always true? **[1]**

a) $\vec{a} \cdot \vec{b} < |\vec{a}||\vec{b}|$ b) $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$

c) $\vec{a} \cdot \vec{b} \le |\vec{a}||\vec{b}|$ d) $\vec{a} \cdot \vec{b} \ge |\vec{a}||\vec{b}|$

13. An ordered pair (α, β) for which the system of linear equations **[1]**

$(1 + \alpha)x + \beta y + z = 2$

$\alpha x + (1 + \beta)y + z = 3$

$\alpha x + \beta y + 2z = 2$

has a unique solution is

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a) (2, 4)

b) (-4, 2)

c) (-3, 1)

d) (1, -3)

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